AWS S3 CLOUD STORAGE SECURITY

Music Uploads Bucket – Practical Assignment

Student Name	Rudra Solanki
Cloud Provider	Amazon Web Services (AWS)
Service	Amazon S3 + IAM
AWS Region	Asia Pacific (Mumbai) – ap-south-1
Bucket Used	my-music-uploads-rudra-2026
IAM User	music-download-user
Date	03 October 2026

Note: The assignment requested the bucket name *my-music-uploads*. Because S3 bucket names are globally unique, the practical implementation uses **my-music-uploads-rudra-2026**.

This report documents the bucket creation, MP3 upload, IAM-controlled download access, server-side encryption, unauthorized browser test, IP-based restriction, and authorized download verification.

1. Objective

The objective of this practical is to create a secure Amazon S3 storage bucket for music files and demonstrate controlled access to the stored objects. The practical covers:

• Create an S3 bucket for music uploads.

• Upload two MP3 files.

• Create an IAM user for controlled download access.

• Apply a bucket policy that permits the designated IAM principal to read objects.

• Enable default server-side encryption using SSE-S3.

• Test that a direct object URL is denied to an unauthenticated browser.

• Apply an IP-based access restriction in the bucket policy.

• Verify that the authorized IAM user can download an MP3 using AWS CLI.

2. Environment and Resources

Component	Value
Cloud	AWS
Storage service	Amazon S3
Region	ap-south-1 (Mumbai)
Bucket	my-music-uploads-rudra-2026
IAM user	music-download-user
Encryption	SSE-S3 (Amazon S3 managed keys)
Objects	2 MP3 files

3. Create the S3 Bucket

A general-purpose S3 bucket was created in the Mumbai region. The console confirms successful creation of **my-music-uploads-rudra-2026**.

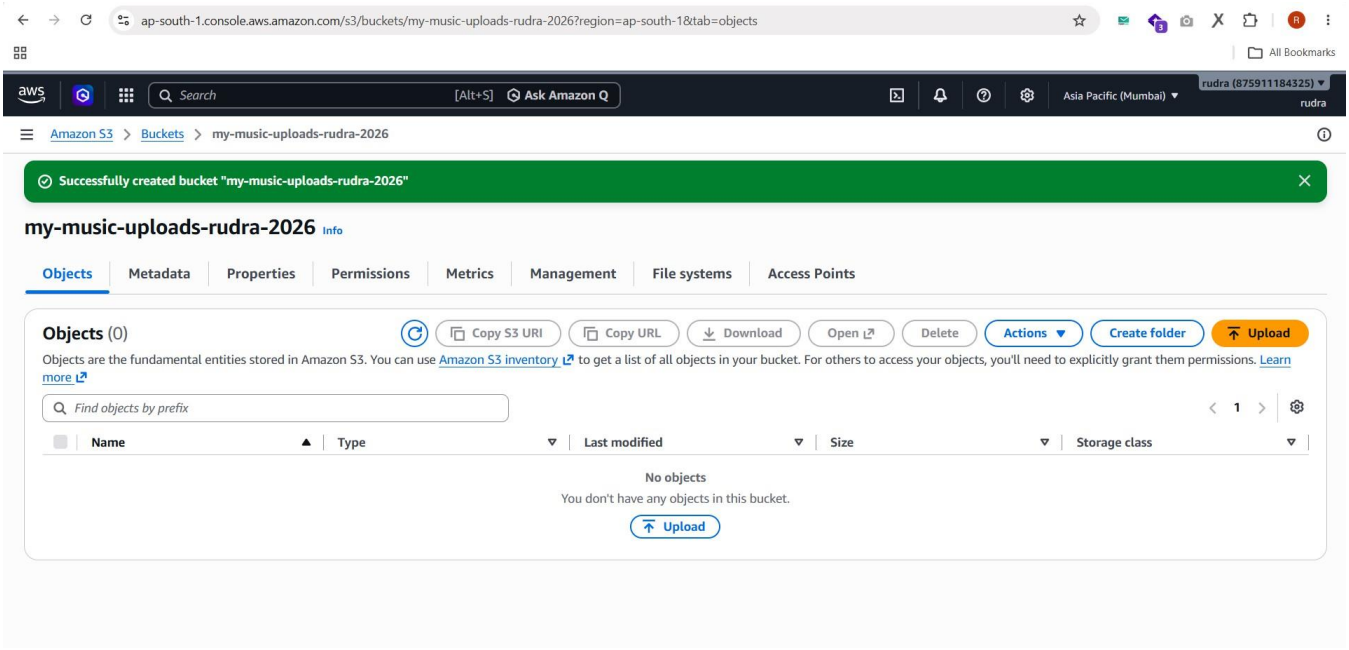


Figure 1 – S3 bucket successfully created.

The bucket is visible in the S3 console and is configured as a general-purpose bucket. The actual name includes a unique suffix because S3 bucket names must be globally unique.

4. Upload Two MP3 Files

Two MP3 files were uploaded successfully. The upload summary reports **2 files**, with a total size of approximately **11.3 MB** and no failed uploads.

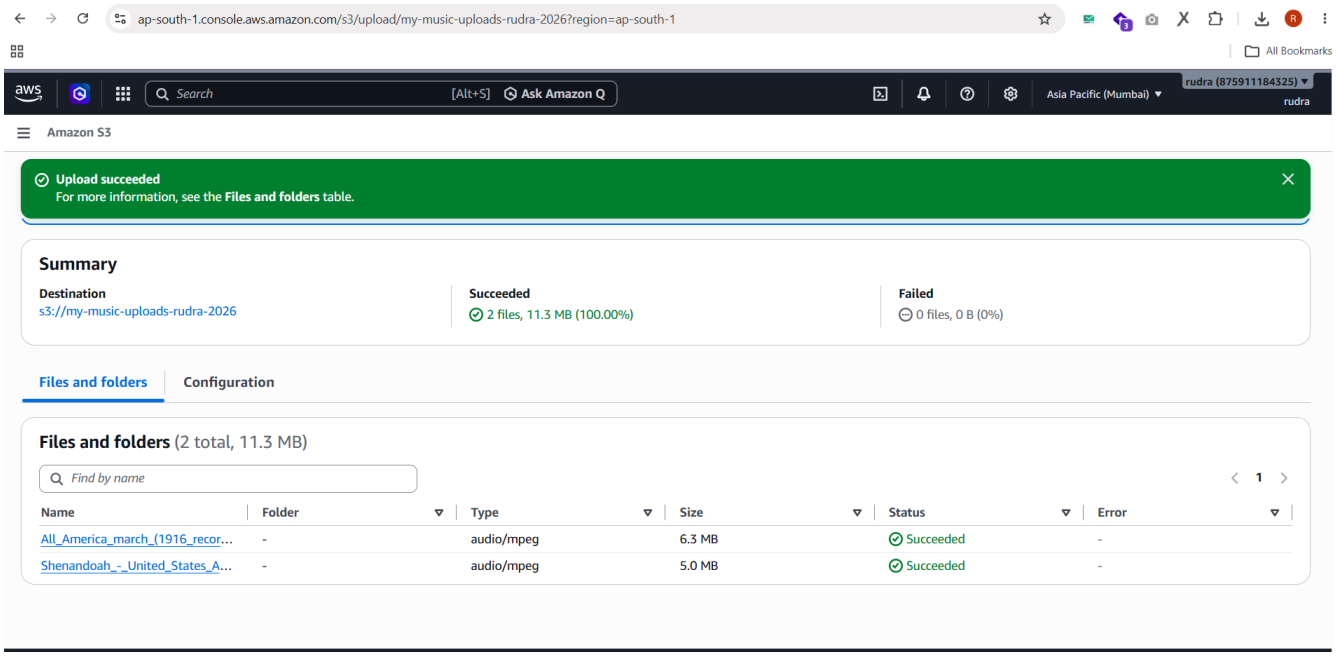


Figure 2 – Upload operation succeeded for two MP3 files.

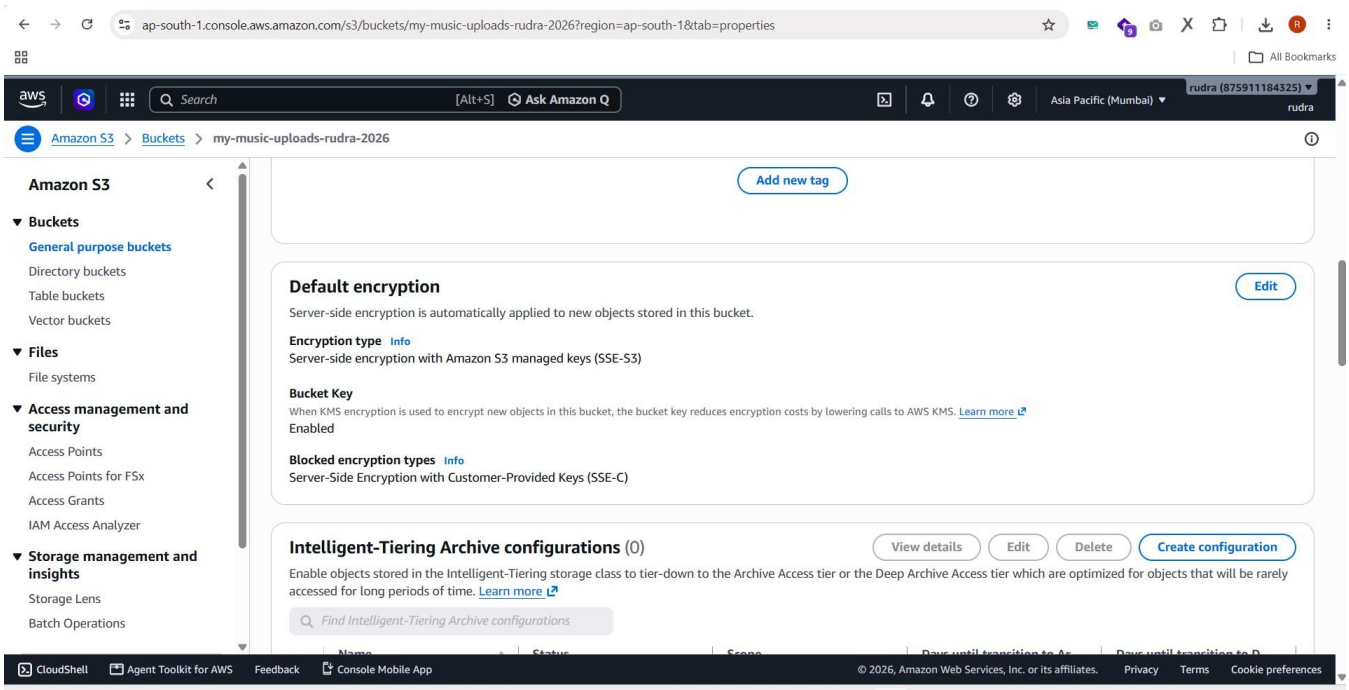


Figure 3 – S3 bucket object list showing the two MP3 files.

5. Create an IAM User for Controlled Downloads

An IAM user named **music-download-user** was created to provide an authenticated identity for downloading objects. Console access was disabled; the user was intended for programmatic S3 access.

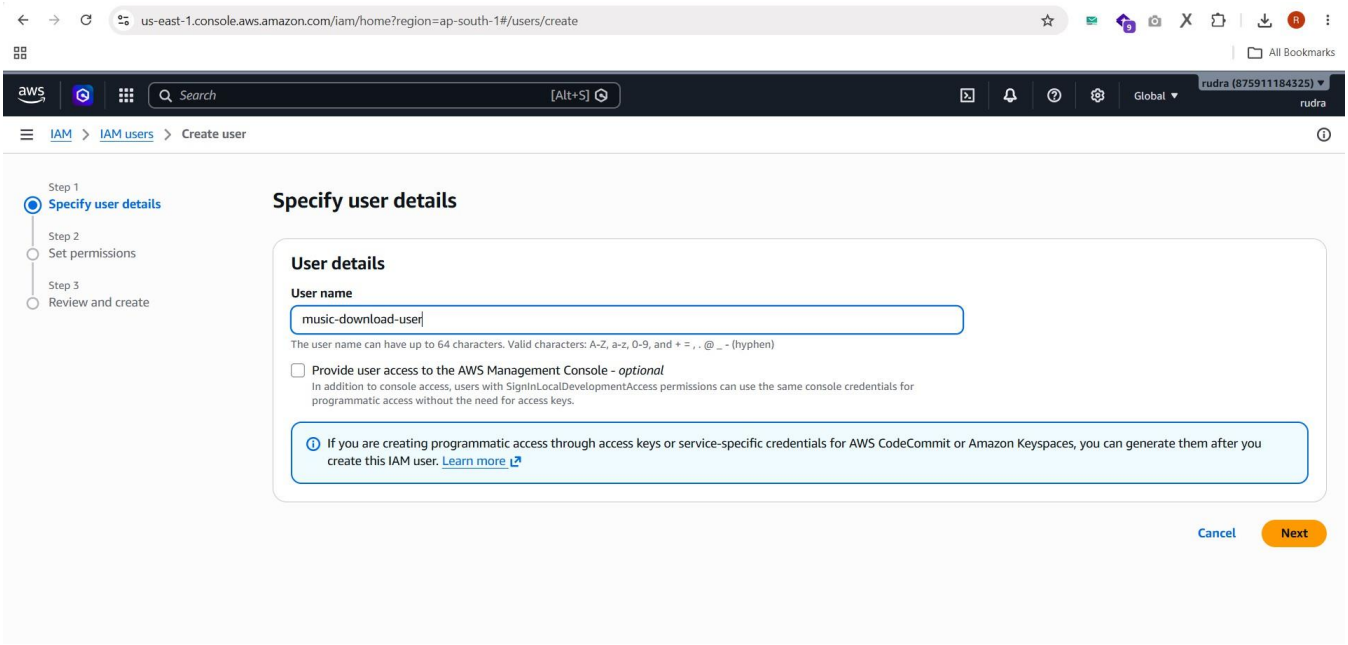


Figure 4 – IAM user creation screen for music-download-user.

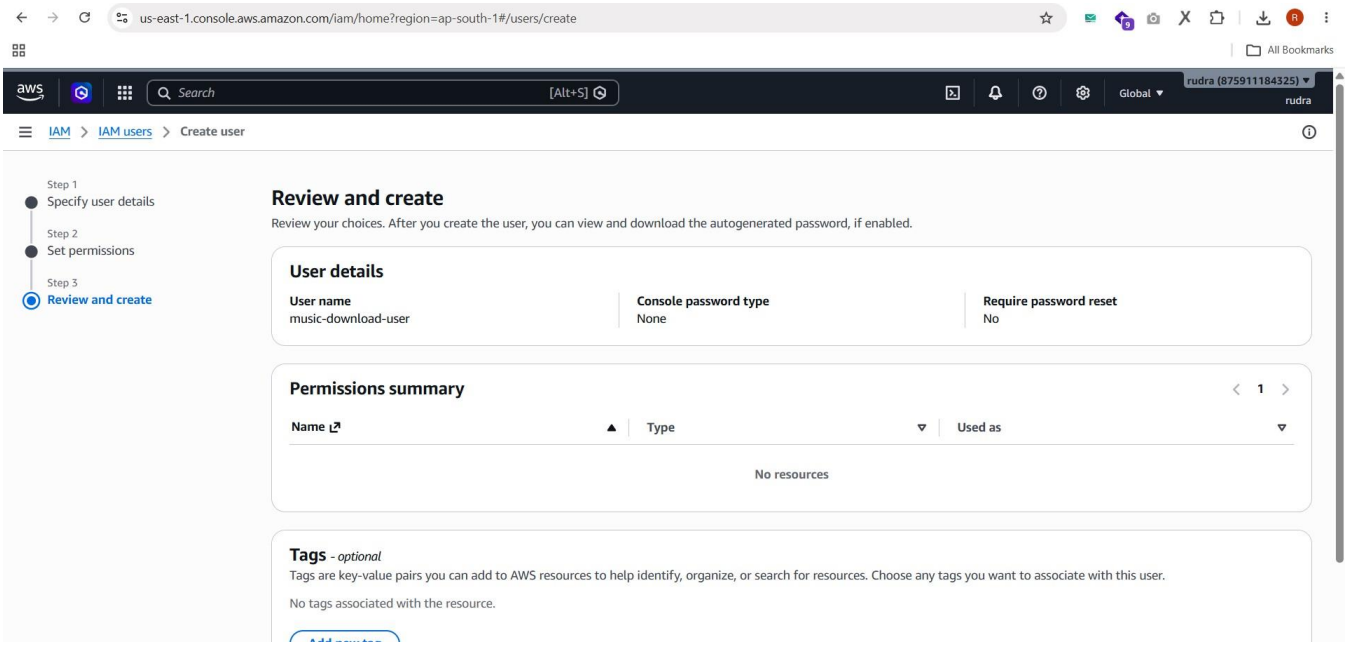


Figure 5 – IAM user review and creation screen.

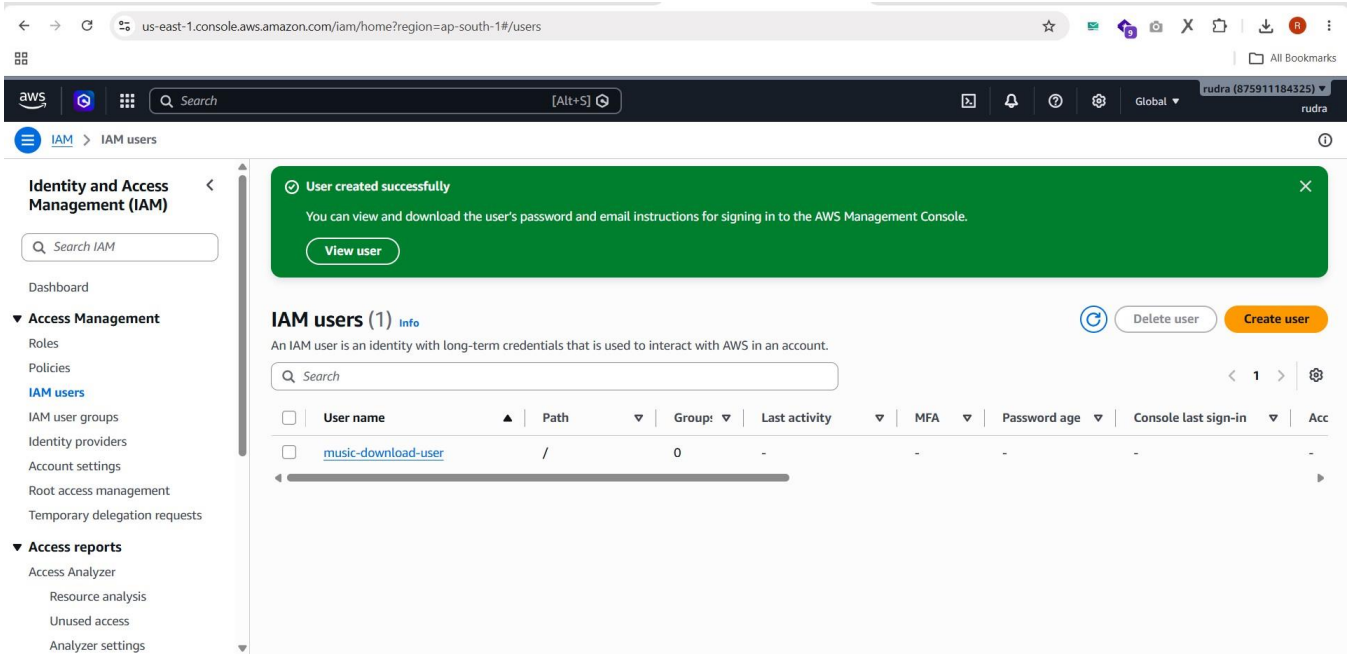


Figure 6 – IAM users list showing music-download-user.

6. IAM Permission Policy

The IAM user has a customer-managed policy named **MusicDownloadPolicy**, attached directly to the user. This policy provides the S3 permissions needed for the practical.

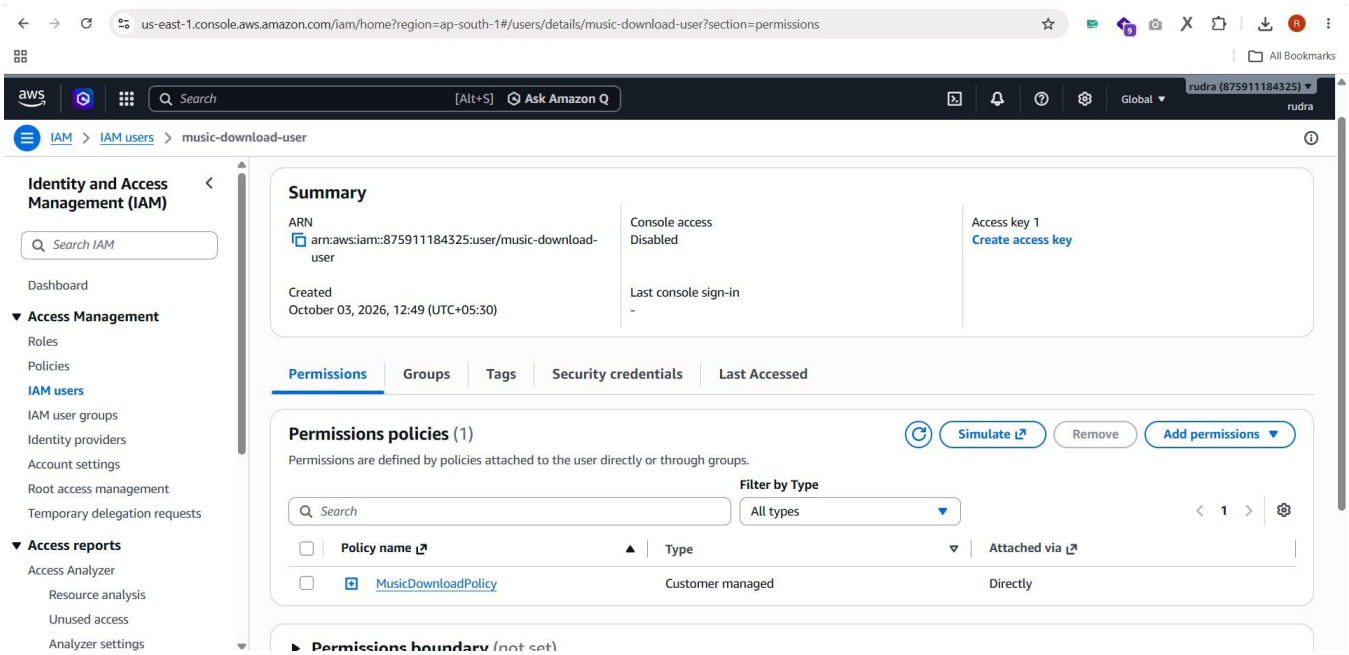


Figure 7 – Permissions for music-download-user showing MusicDownloadPolicy.

Sample policy structure used for the download permission:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "s3:GetObject",
      "Resource":  "arn:aws:s3:::my-music-uploads-rudra-2026/*"
    }
  ]
}
```

This grants the IAM identity permission to retrieve objects from the bucket. The bucket policy shown in the next section further controls which principal and network conditions are allowed.

7. Bucket Policy and IP Restriction

The bucket policy was successfully edited. The visible policy contains an Allow statement for the **music-download-user** IAM principal and a Deny statement named **DenyRequestsOutsideAllowedIP** using an **aws:NotIpAddress** condition. This implements an IP-based restriction for object downloads.

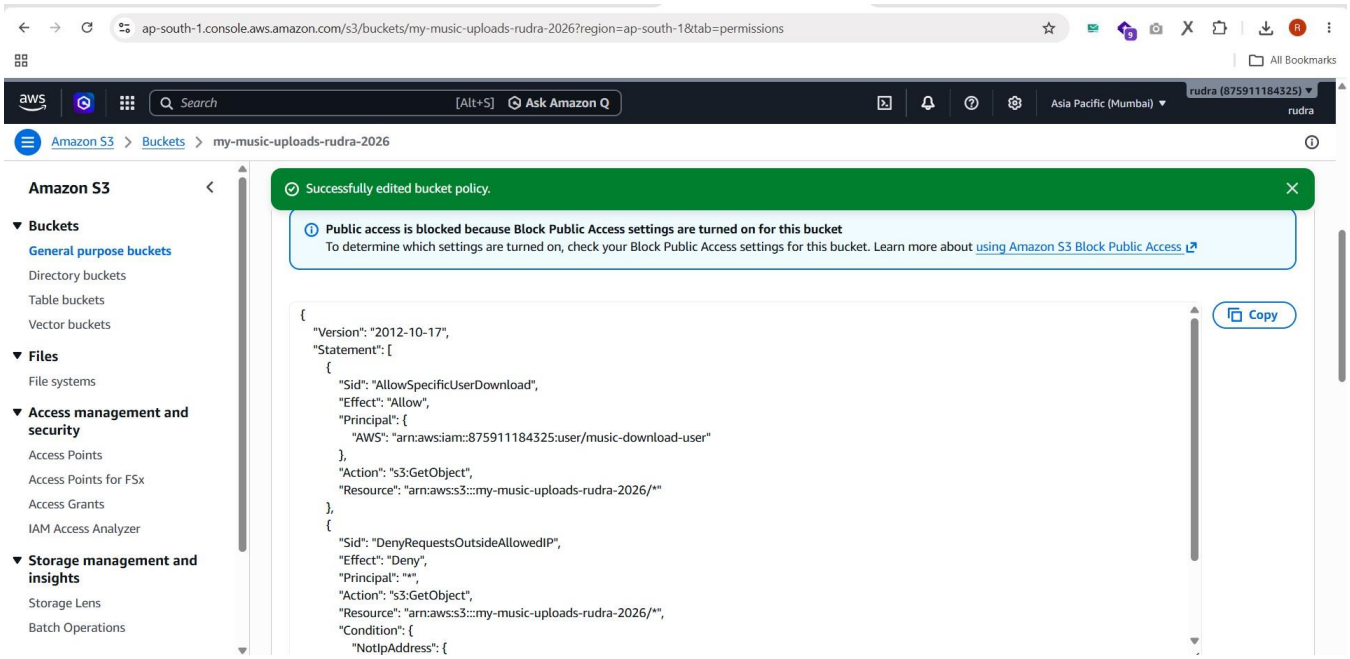


Figure 8 – Bucket policy successfully edited; the policy includes an IAM allow and an IP-based deny.

Representative IP-restriction policy pattern:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyRequestsOutsideAllowedIP",
      "Effect": "Deny",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-music-uploads-rudra-2026/*",
      "Condition": {
        "NotIpAddress": {
          "aws:SourceIp": "/32"
        }
      }
    }
  ]
}
```

The placeholder above represents the public IP that was configured for the practical. The screenshot confirms that the IP-deny statement was applied, but the exact IP value is not visible in the captured portion of the console.

8. Enable Server-Side Encryption (SSE-S3)

Default server-side encryption was enabled for the bucket using **SSE-S3**, which uses Amazon S3 managed encryption keys. The console states that new objects stored in the bucket automatically inherit this encryption configuration unless otherwise specified.

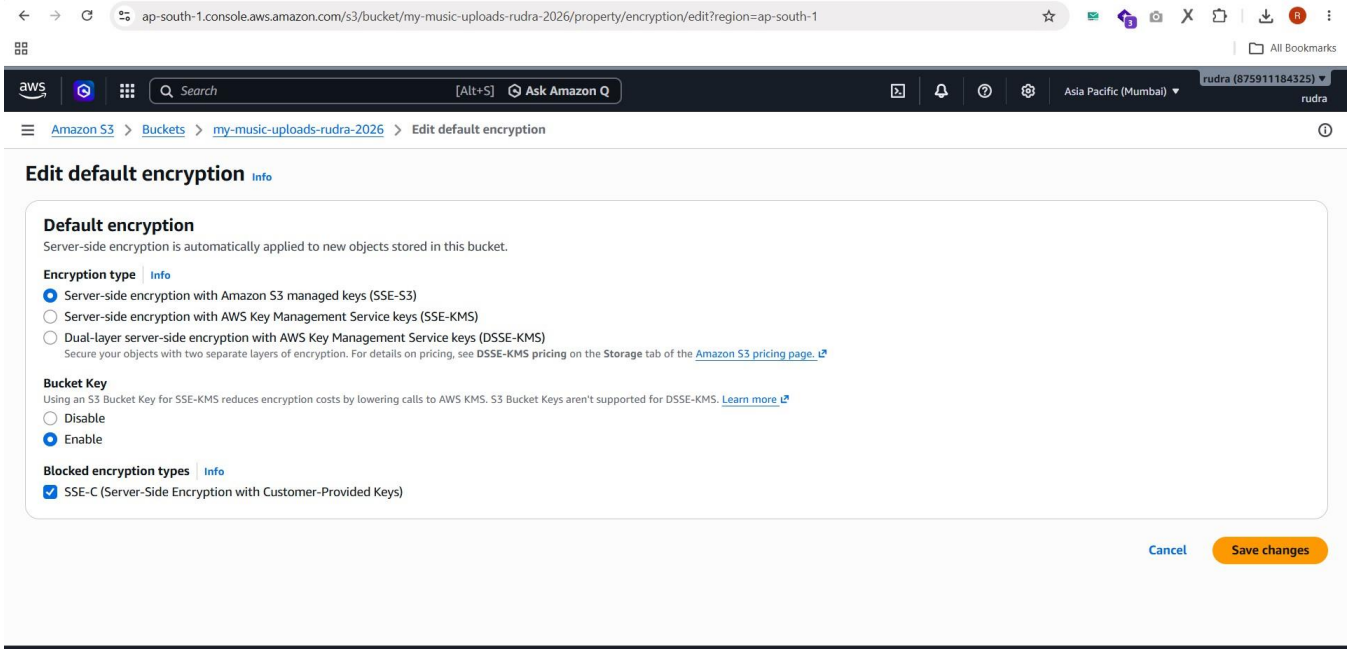


Figure 9 – Default encryption configured as SSE-S3.

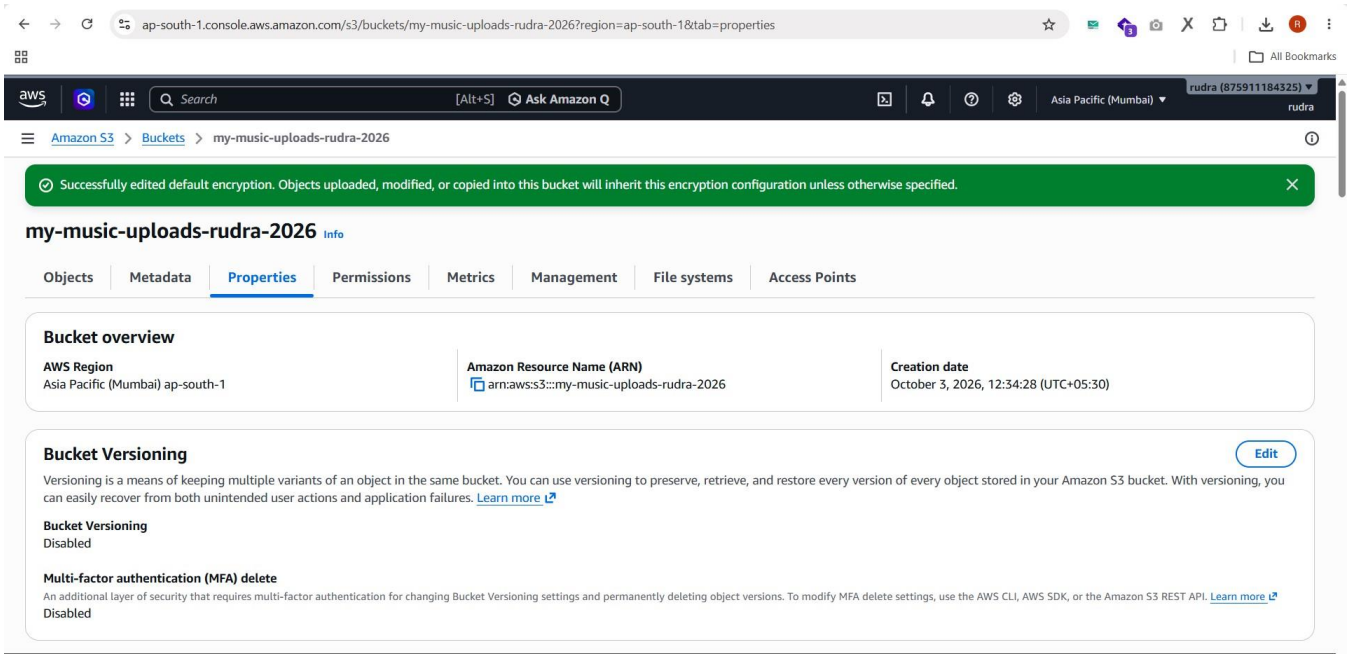


Figure 10 – Successful encryption configuration update.

```
Configure AWS skills and the AWS MCP server for your AI coding agent(s)? [y/n/never]: n
PS C:\Users\rudra> aws sts get-caller-identity
{
    "UserId": "AIDA4X4C777C4MDH46IX2",
    "Account": "875911184325",
    "Arn": "arn:aws:iam::875911184325:user/music-download-user"
}

PS C:\Users\rudra>
```

Figure 11 – Bucket Properties showing default SSE-S3 encryption enabled.

9. Verify an Uploaded MP3 Object

The object details page confirms the MP3 object exists in the bucket, including its S3 URI, object ARN, size, type, and region. This provides object-level evidence that the uploaded media is stored in S3.

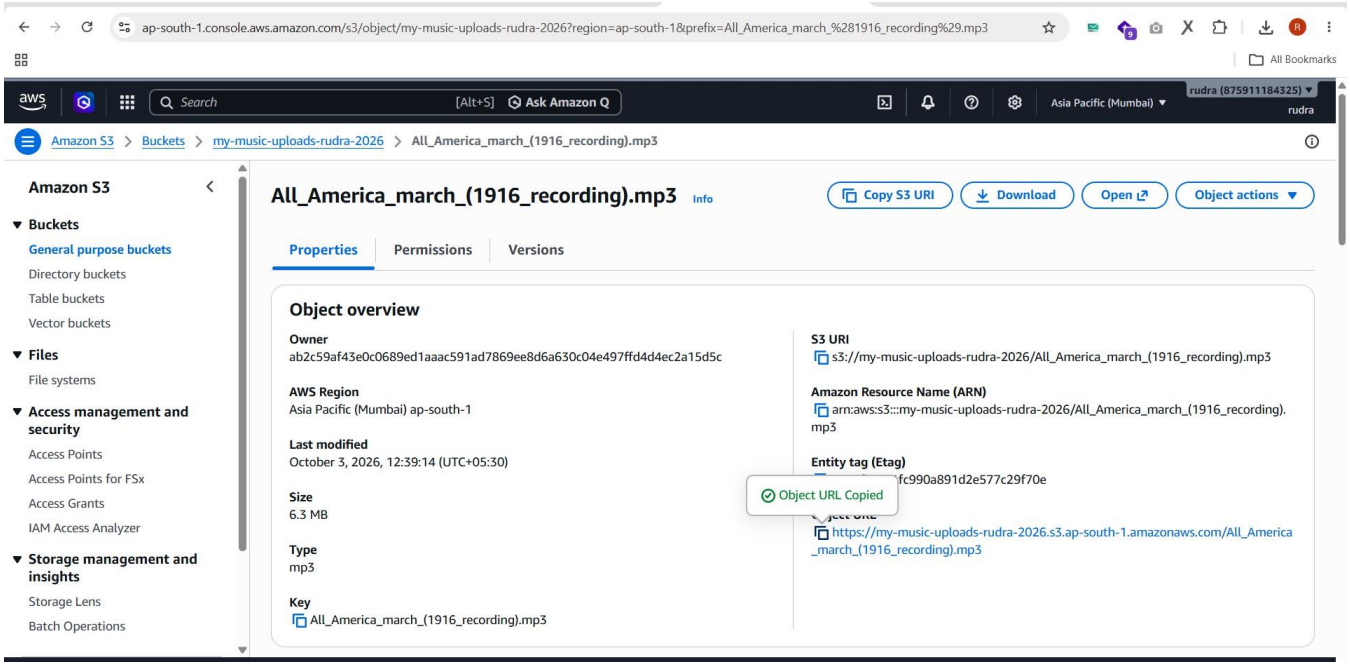


Figure 12 – Uploaded MP3 object properties and S3 URI.

10. Test Unauthorized Browser Access

The S3 object URL was opened directly in a browser without an authenticated AWS session. The browser returned an XML error containing **AccessDenied**. This demonstrates that the object cannot be downloaded anonymously through the direct URL.

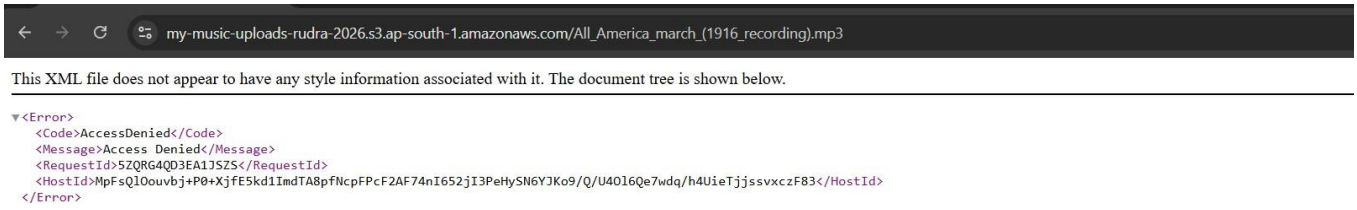


Figure 13 – Unauthenticated browser request denied with AccessDenied.

11. Verify Authenticated IAM Access

AWS CLI was used with the IAM user's credentials. The **aws sts get-caller-identity** command confirms that the active identity is the IAM user **music-download-user**.

```
Configure AWS skills and the AWS MCP server for your AI coding agent(s)? [y/n/never]: n
PS C:\Users\rudra> aws sts get-caller-identity
{
    "UserId": "AIDA4X4C777C4MDH46IX2",
    "Account": "875911184325",
    "Arn": "arn:aws:iam::875911184325:user/music-download-user"
}

PS C:\Users\rudra>
```

Figure 14 – Bucket encryption settings (supporting security configuration).

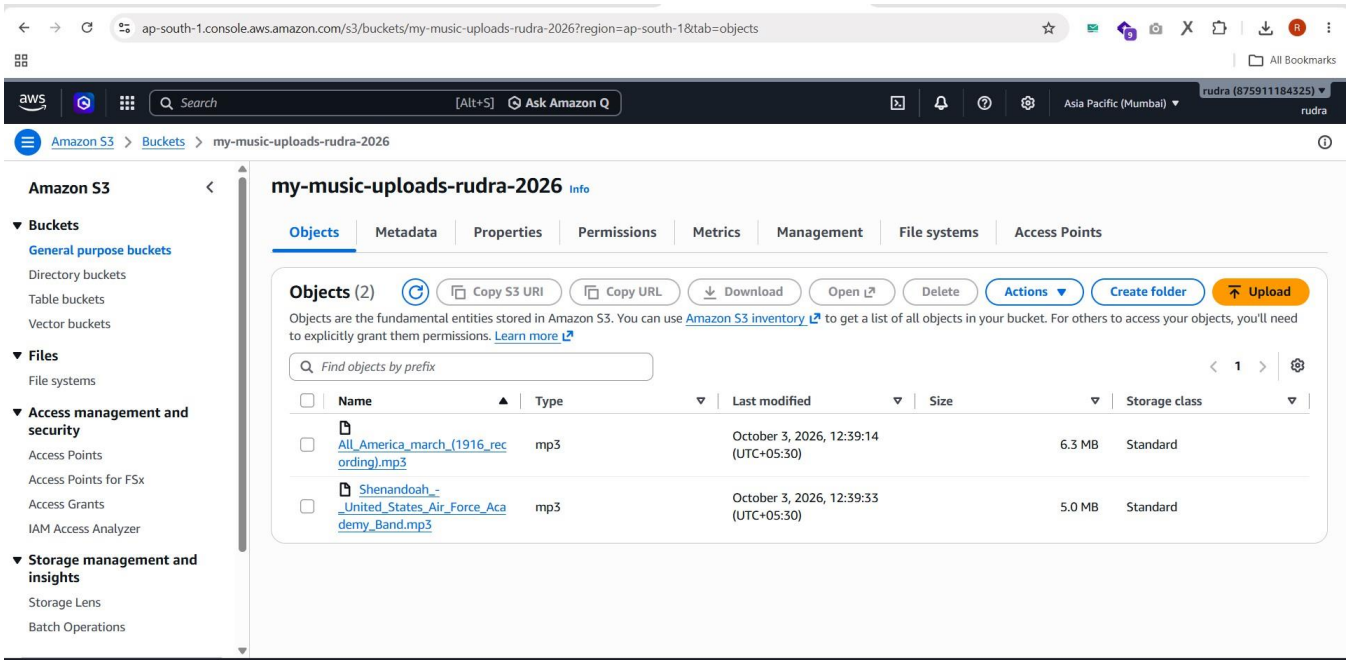


Figure 15 – AWS CLI get-caller-identity confirms the music-download-user identity.

12. Authorized IAM User Download

The authenticated IAM user successfully downloaded the MP3 from S3 using the AWS CLI. The command output reports the S3 object was downloaded to the local Windows directory.

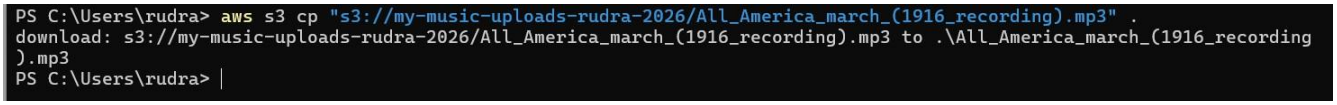


Figure 16 – AWS CLI download succeeds for the MP3 object.

This completes the access-control test: anonymous browser access was denied, while an authenticated IAM identity with the required S3 permission was able to download the object.

13. Final Verification / Results

Requirement	Evidence / Result
S3 bucket created	my-music-uploads-rudra-2026 created successfully.
Two MP3 files uploaded	2 objects are visible in the bucket; upload summary shows 2 successful files.
IAM download identity	music-download-user created and has MusicDownloadPolicy.
Bucket access control	Bucket policy includes an Allow for the IAM principal.
IP restriction	DenyRequestsOutsideAllowedIP uses aws:NotIpAddress.
Server-side encryption	Default SSE-S3 is enabled for new objects.
Anonymous browser test	Direct object URL returned AccessDenied.
Authenticated download	AWS CLI successfully downloaded the MP3 as music-download-user.

14. Conclusion

The AWS S3 practical was completed using the bucket **my-music-uploads-rudra-2026**. Two MP3 files were uploaded and protected using IAM-based permissions, an IP-based bucket restriction, and default SSE-S3 encryption. The security test showed that an unauthenticated browser request received **AccessDenied**, while the authorized IAM user was able to download an object through AWS CLI. The practical therefore demonstrates controlled access, encryption at rest, and network-based restriction for objects stored in Amazon S3.